

Antenna Datasheet

Product OC: YECT001WAAM
Version: 1.0
Date: 2026-06-08
Status: Released

Product Name: 4G Terminal Mount Rubber Monopole External Antenna
Key Features:
Frequency Band: 700–960 MHz, 1710–2690 MHz
Dimensions: Φ 14.5 mm × 57.5 mm
Efficiency: Up to 52.0 %
RoHS Compliant
IP68

Overview

YECT001WAAM is a 4G rubber antenna measuring Φ 14.5 mm × 57.5 mm. This ultra-wide-band 4G antenna provides broad coverage from 700–960 MHz, 1710–2690 MHz whilst offering backward-compatibility to support 3G and 2G networks as well as LTE Cat-M and narrowband IoT (NB-IoT). The antenna is terminated with TNC Male connector. This low profile, terminal mount omni-directional antenna, ideal for applications where the antenna is required to be discrete, is easy to install with maximum durability assured thanks to its TPE enclosure. It is compatible with Quectel 's 4G Series modules.

It allows constant and reliable transmission and reception due to its omni-directional gain across all frequency bands. It is a perfect antenna product for customers that desire highest performance. This high-efficiency, high-gain omni-directional antenna is ideally suited for Zigbee, Bluetooth, IoT Sensors, public safety and security, point of sales terminals, smart home automation, robotics / autonomous.

- **Typical applications include:**

✓ Zigbee
✓ Bluetooth
✓ IoT Sensors
✓ Public Safety and Security
✓ Point of Sales Terminals
✓ Smart Home Automation
✓ Robotics / Autonomous

Quectel provides comprehensive antenna design support such as simulation, testing and manufacturing for custom antenna solutions to meet your specific application needs. We have regional R & D centers to offer quick response to meet your requirements. Please contact our sales & FAEs if you have any requests.

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1 Specification

Test Condition: Free Space

1.1. Electrical

Electrical	
Frequency Range	700–960 MHz, 1710–2690 MHz
Impedance	50 Ω
Polarization	Linear
Radiation Pattern	Omni-directional
Antenna Type	Monopole

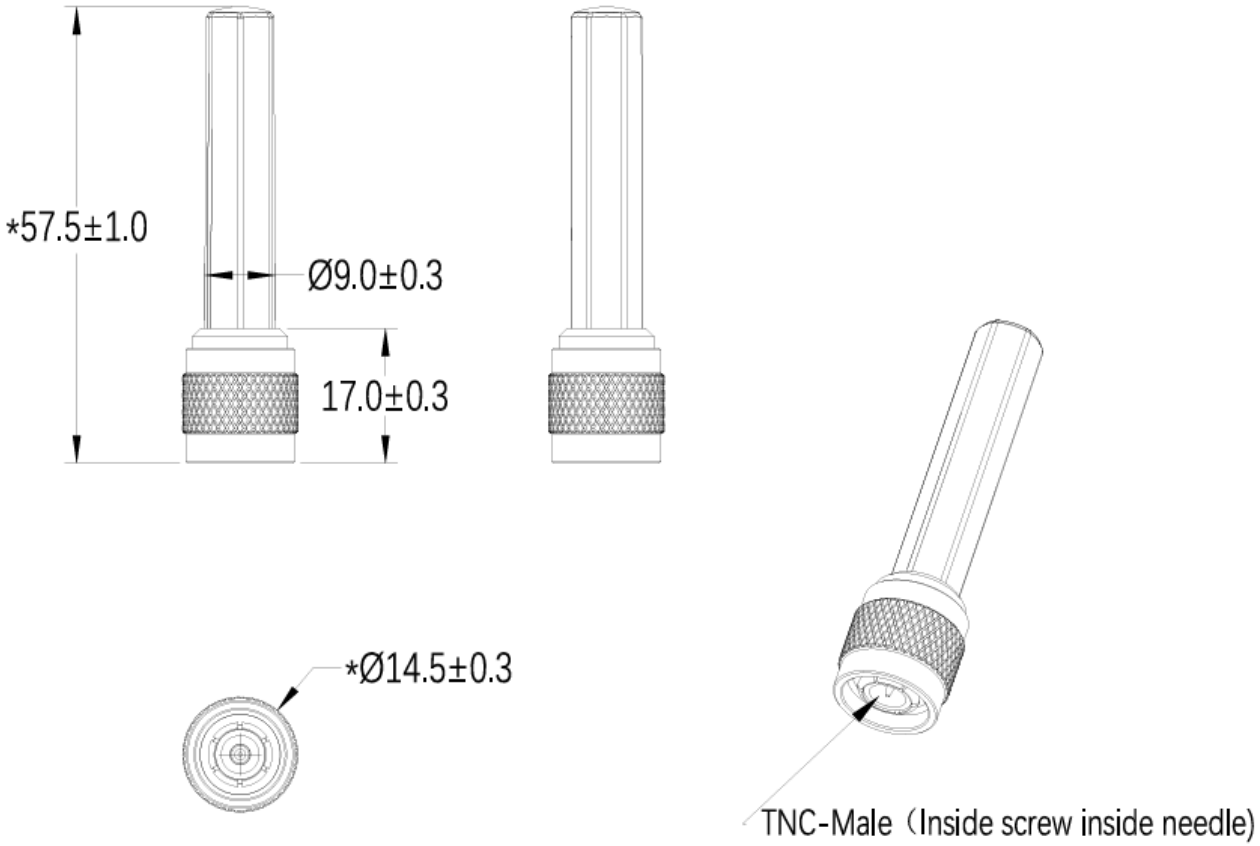
Electrical – Detail

Band / SPEC	Band	B71	B12 /B13 /B28	B5 /B8 /B26	n74 /n75 /n76	B1 /B2 /B3	B40	Wi-Fi 2G	B38 /B41	B42 /B48 /n77	n79	Wi-Fi 5G
	Freq. (MHz)	600–700	700–810	820–960	1420–1520	1700–2170	2300–2400	2400–2500	2500–2690	3300–4200	4400–5000	5150–5850
Max. VSWR	-	14.3	8.6	-	4.6	2.6	3.1	3.2	-	-	-	
Max. Return Loss (dB)	-	-1.2	-2.0	-	-3.8	-7.1	-5.7	-5.7	-	-	-	
AVG Eff. (%)	-	8.8	16.7	-	39.6	48.4	43.7	45.1	-	-	-	
AVG AVG Gain (dB)	-	-10.6	-7.9	-	-4.0	-3.2	-3.6	-3.5	-	-	-	
Max. Peak Gain (dBi)	-	-5.3 (810)	-0.3 (960)	-	3.3 (2170)	3.6 (2320)	3.2 (2400)	1.9 (2530)	-	-	-	
VSWR	≤ 14.3											
Return Loss	≤ -1.2 dB											
Peak Gain	≤ 3.6 dBi											

1.2. Mechanical & Environmental

Mechanical	
Antenna Dimensions	Φ 14.5 mm × 57.5 mm
Material & Color	TPE & Black
Connector Type	TNC Male
Mounting Type	Terminal
Weight	Typ. 13.5 g
Environmental	
Operation Temperature	-40 °C to +85 °C
Storage Temperature	-40 °C to +85 °C
Ingress Protection (IP) Rating	IP68
RoHS Compliant	Yes

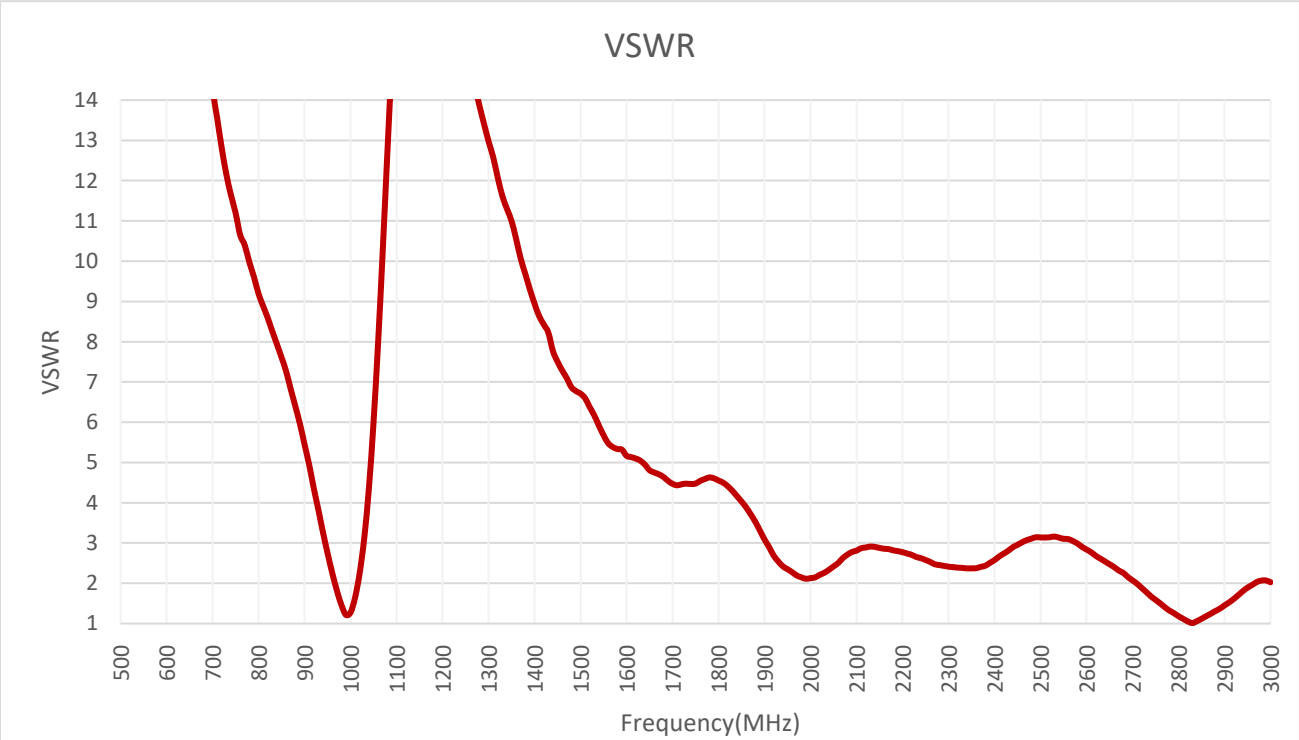
2 Drawing


*57.5±1.0
Ø9.0±0.3
17.0±0.3
*Ø14.5±0.3
TNC-Male（Inside screw inside needle)

3 Detailed Performance

3.1. S-Parameter Test

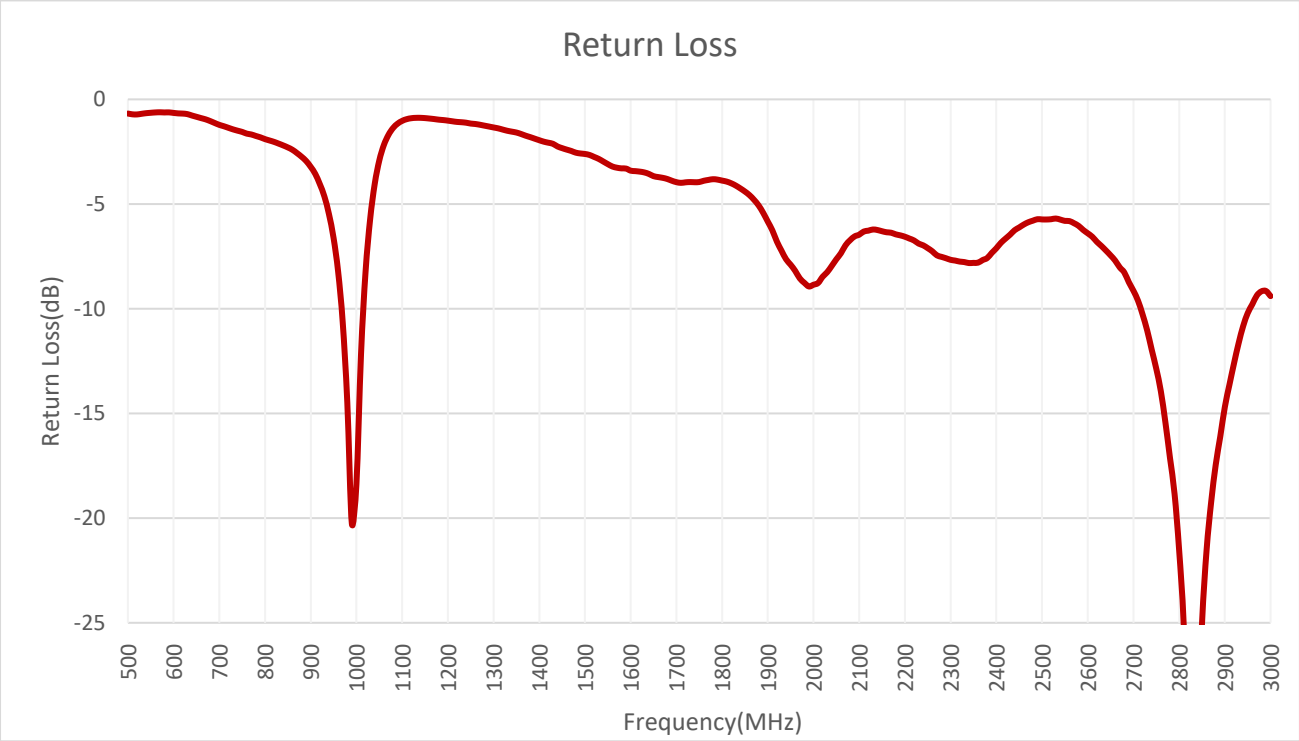
3.1.1. VSWR



VSWR

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
VSWR	-	-	13.6	8.3	5.5	2.3	-	4.4	4.5	3.5
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4700**	**5000**	**5500**	**6000**
VSWR	2.3	2.9	2.4	3.0	2.8	-	-	-	-	-

3.1.2. Return Loss

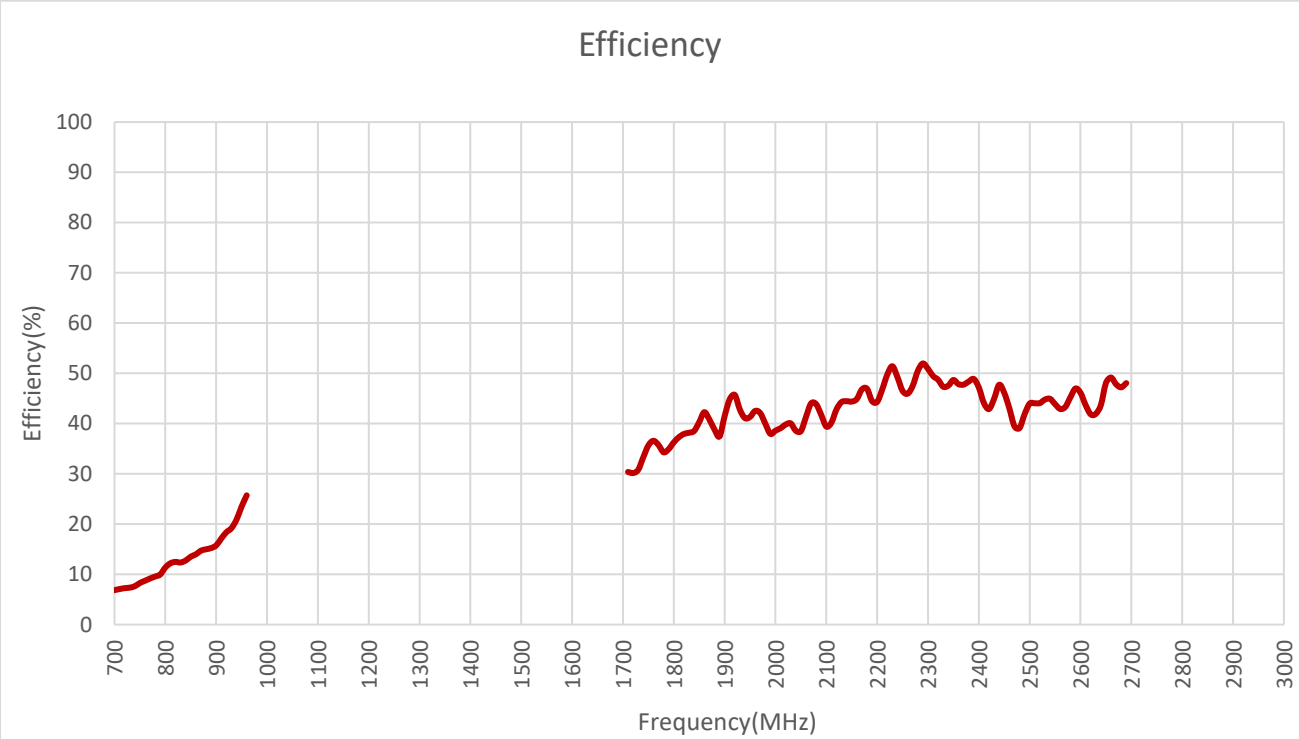


Return Loss (dB)

Frequency (MHz)	**600**	**630**	**710**	**830**	**900**	**960**	**1440**	**1710**	**1740**	**1880**
Return Loss (dB)	-	-	-1.3	-2.1	-3.2	-8.2	-	-4.0	-4.0	-5.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4700**	**5000**	**5500**	**6000**
Return Loss (dB)	-7.9	-6.2	-7.8	-6.1	-6.4	-	-	-	-	-

3.2. Radiation Performance Test

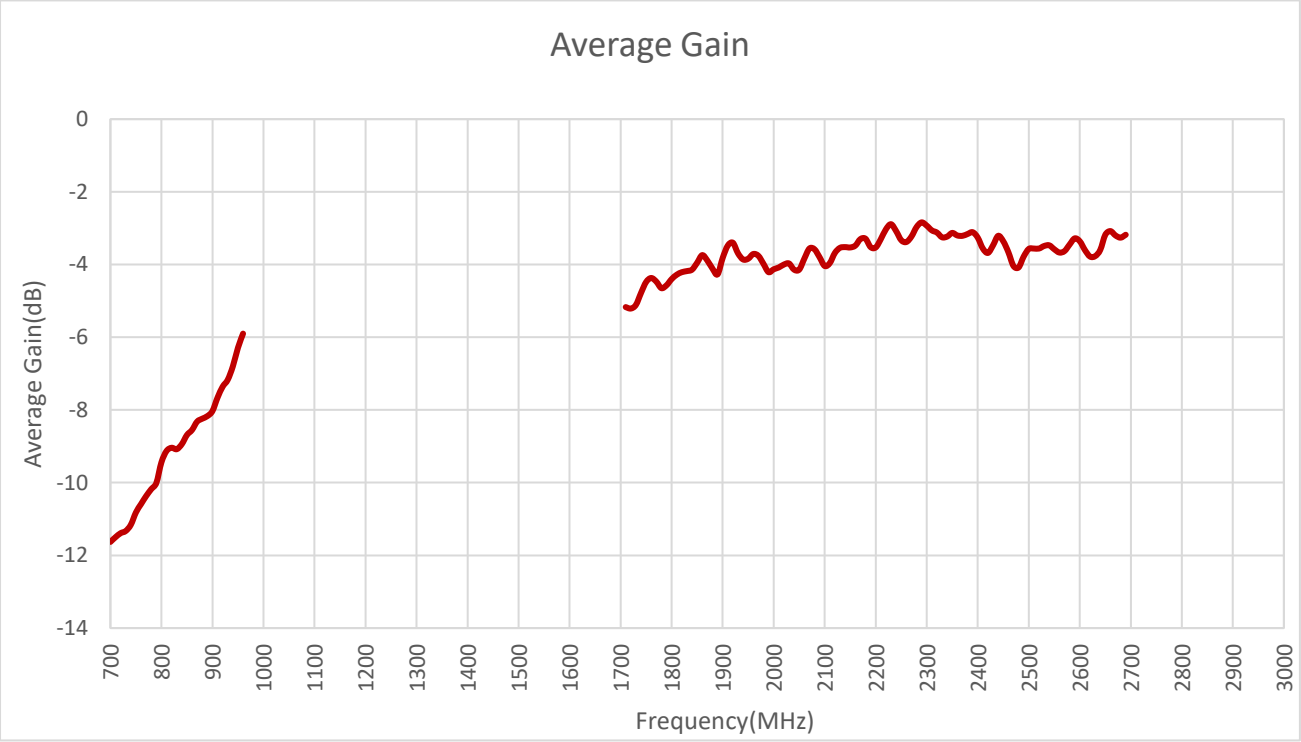
3.2.1. Efficiency



Efficiency (%)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Efficiency (%)	-	-	7.1	12.4	15.8	25.7	-	30.4	33.3	38.8
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4700**	**5000**	**5500**	**6000**
Efficiency (%)	41.3	44.5	48.7	46.1	46.1	-	-	-	-	-

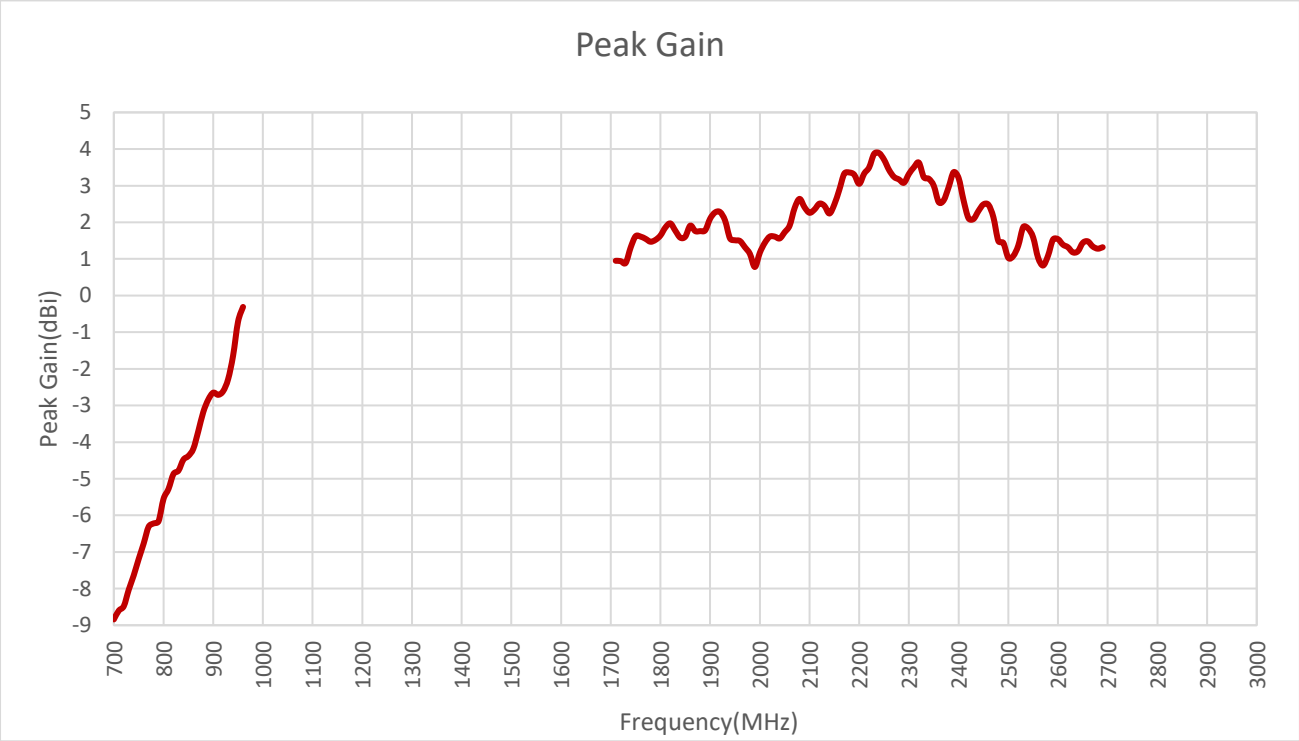
3.2.2. Average Gain



Average Gain (dB)

Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Average Gain (dB)	-	-	-11.5	-9.1	-8.0	-5.9	-	-5.2	-4.8	-4.1
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4700**	**5000**	**5500**	**6000**
Average Gain (dB)	-3.8	-3.5	-3.1	-3.4	-3.4	-	-	-	-	-

3.2.3. Peak Gain

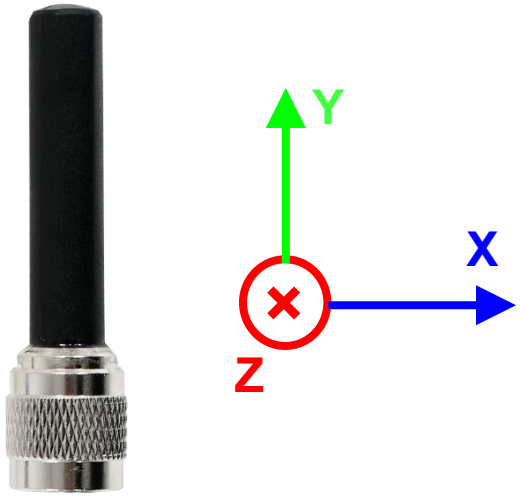


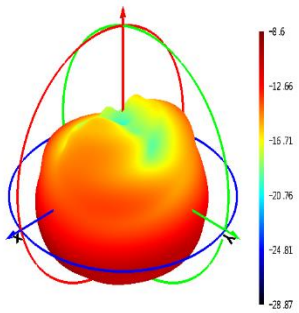
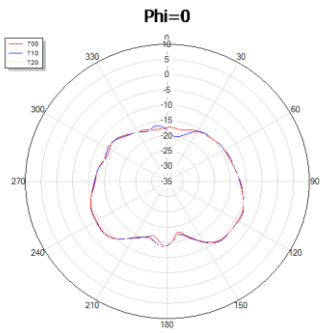
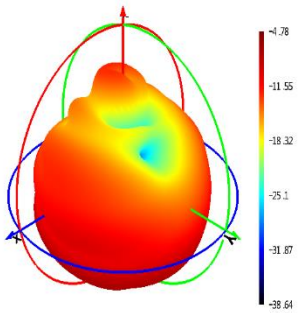
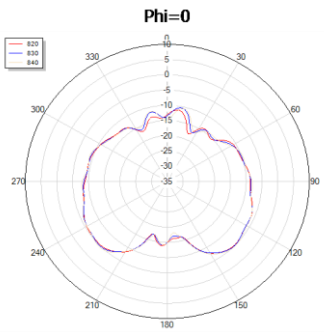
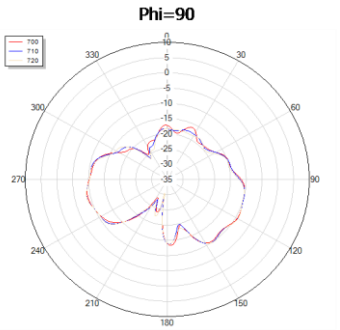
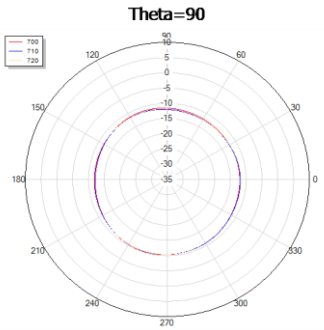
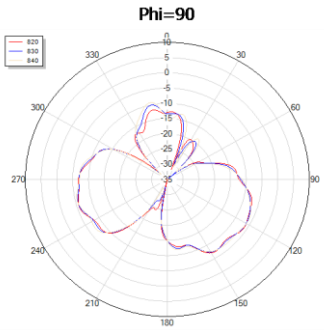
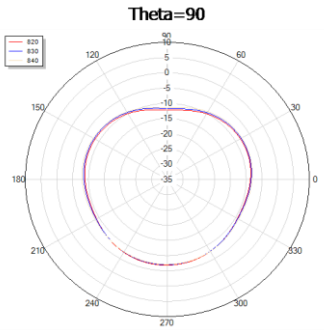
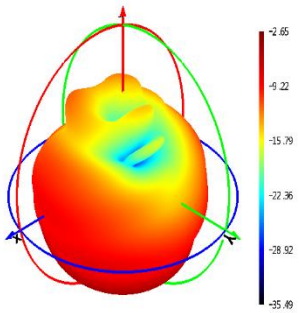
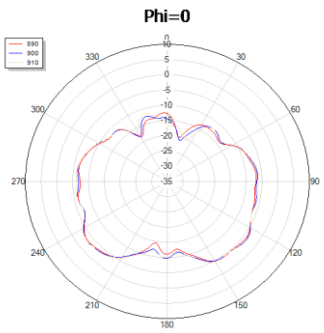
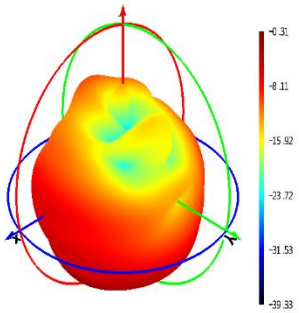
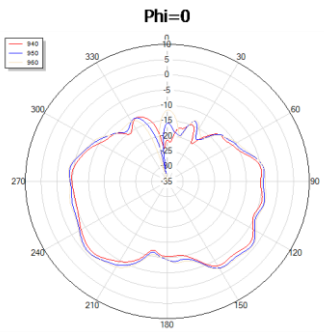
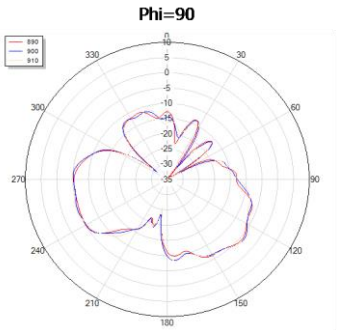
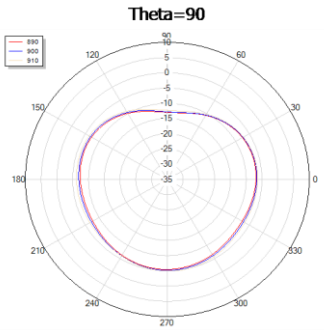
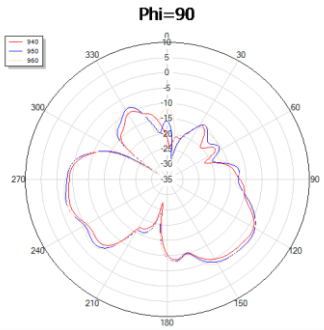
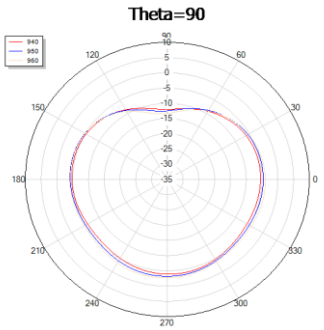
Peak Gain (dBi)

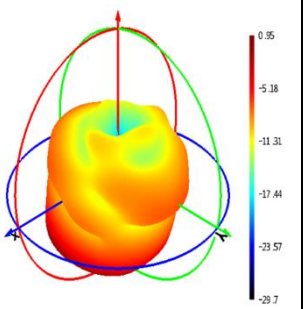
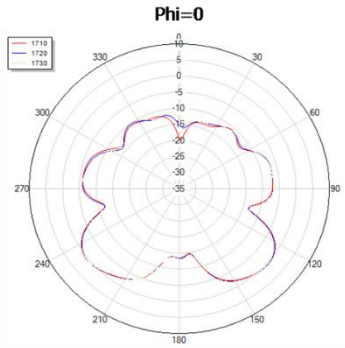
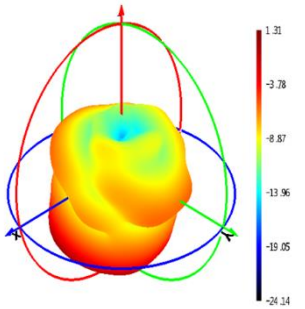
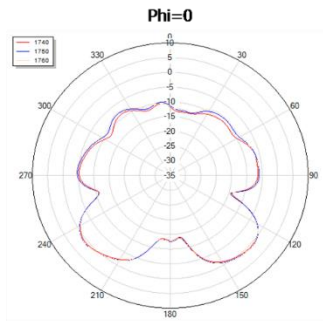
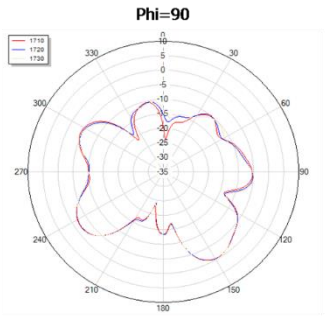
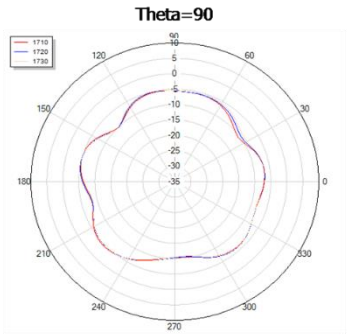
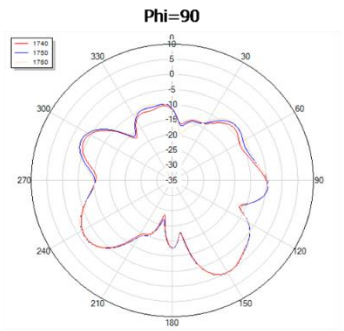
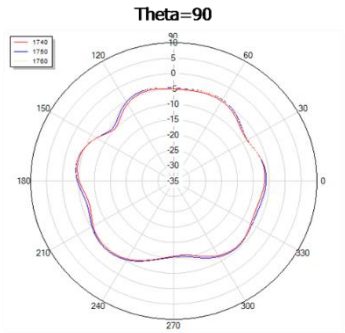
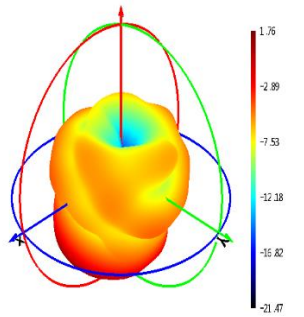
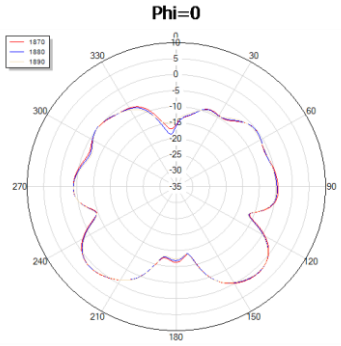
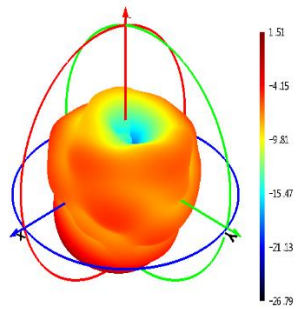
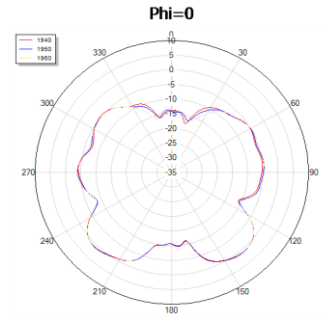
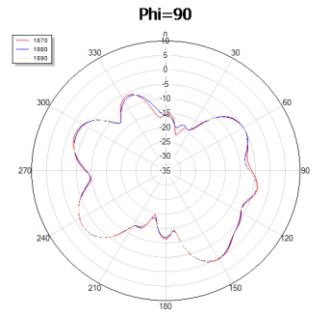
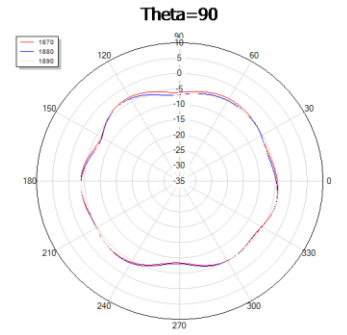
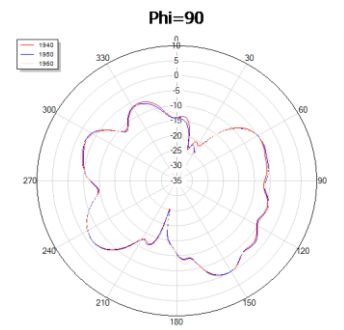
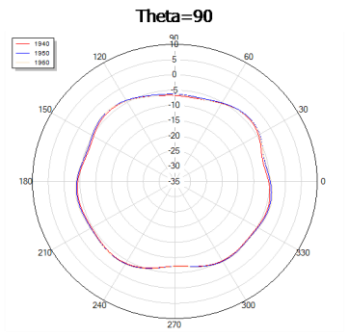
Frequency (MHz)	600	630	710	830	900	960	1440	1710	1740	1880
Peak Gain (dBi)	-	-	-8.6	-4.8	-2.7	-0.3	-	1.0	1.3	1.8
Frequency (MHz)	**1950**	**2140**	**2350**	**2450**	**2600**	**3600**	**4700**	**5000**	**5500**	**6000**
Peak Gain (dBi)	1.5	2.2	3.0	2.5	1.5	-	-	-	-	-

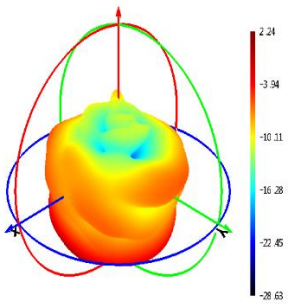
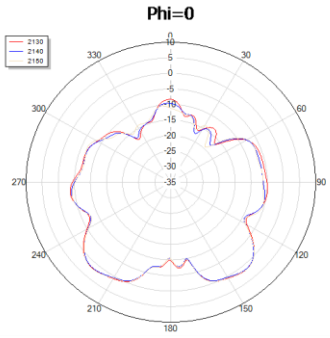
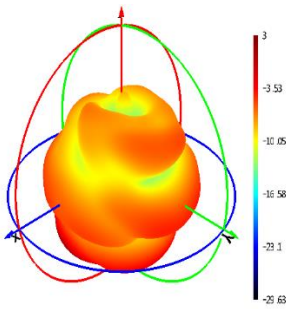
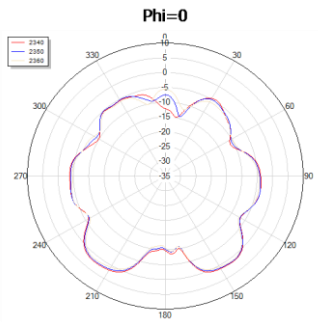
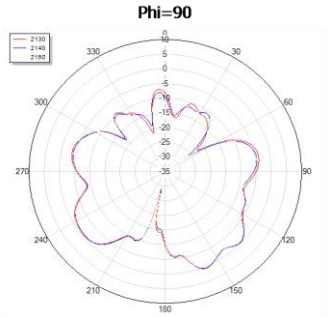
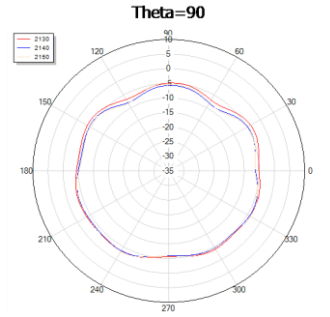
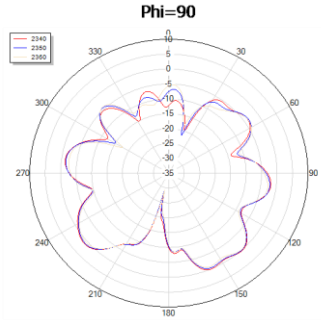
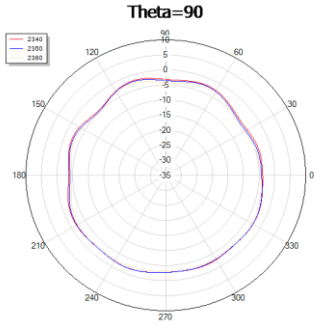
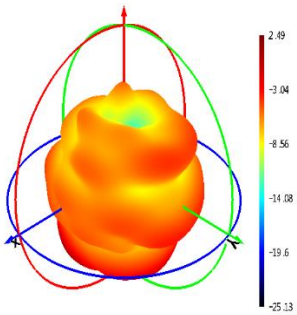
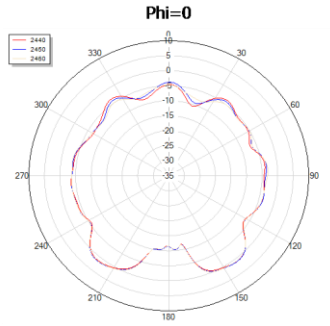
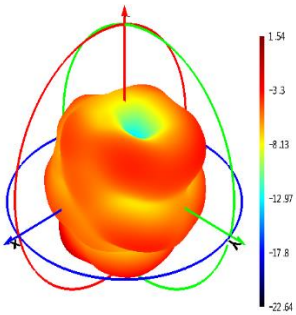
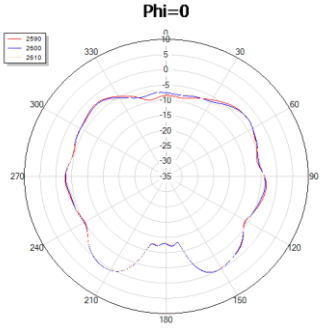
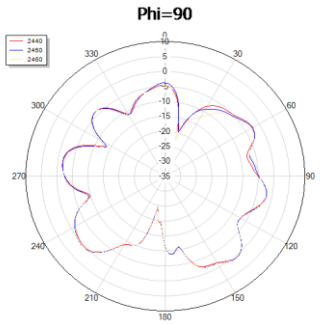
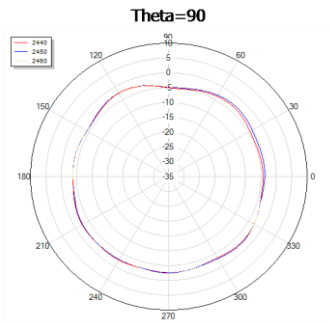
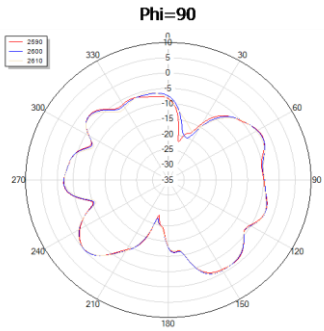
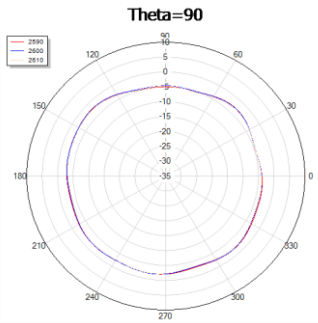
3.2.4. 3D & 2D Radiation Pattern

- Test Condition: Free space
- Test Chamber: HF-S-1

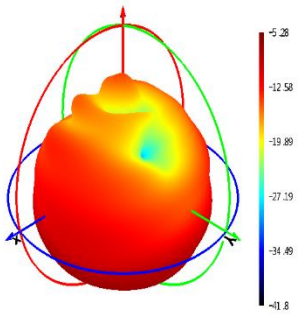
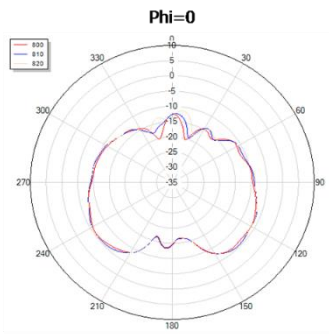
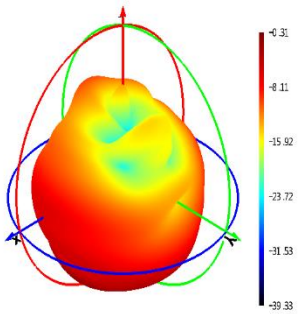
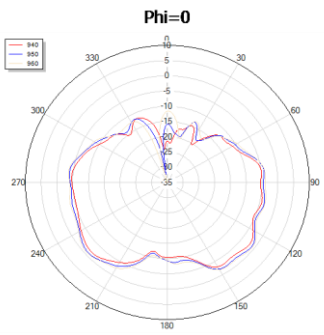
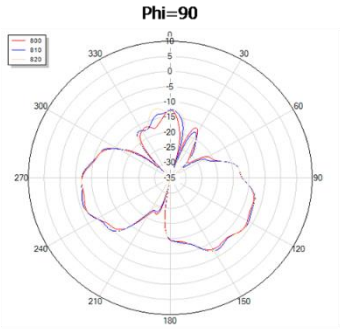
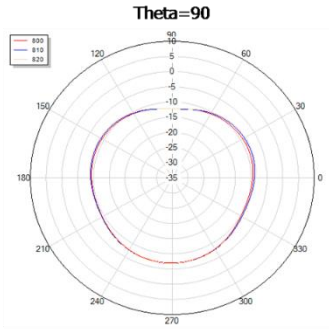
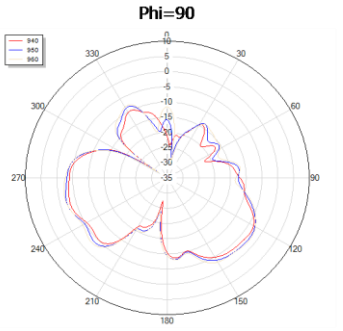
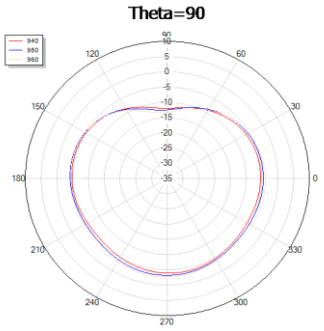
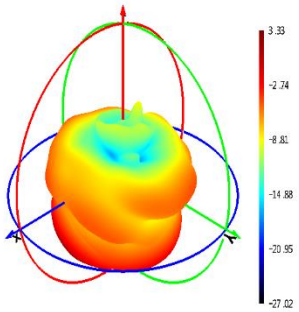
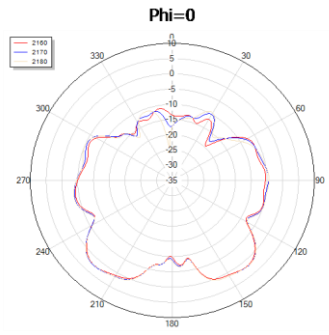
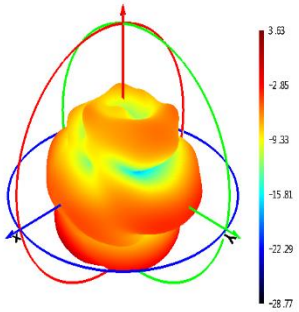
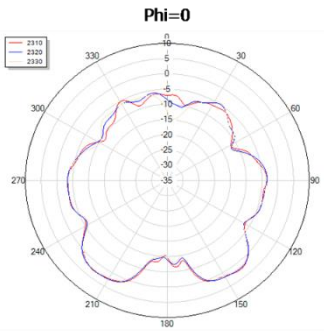
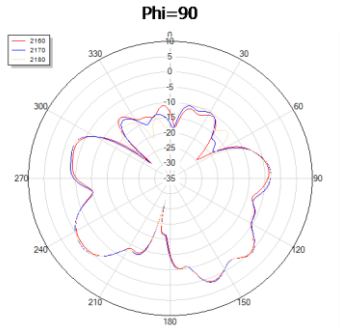
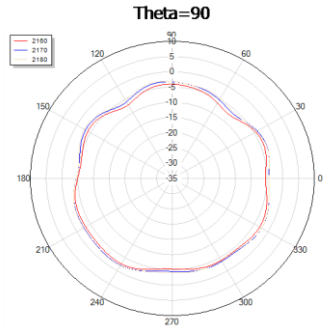
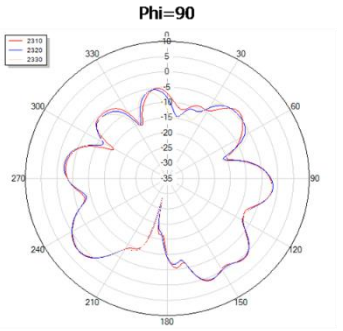
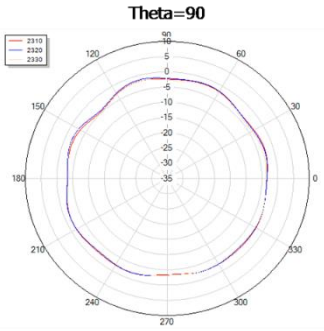


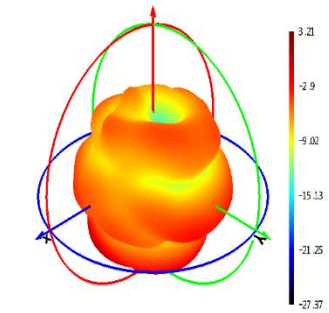
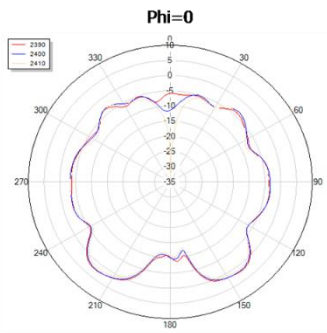
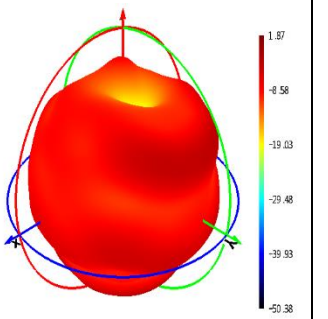
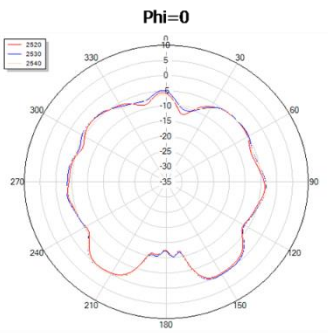
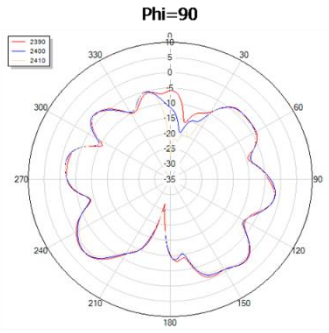
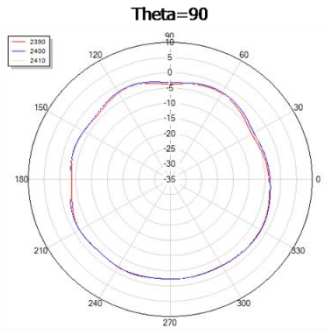
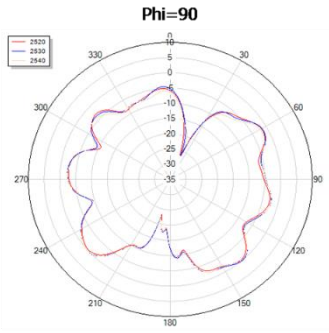
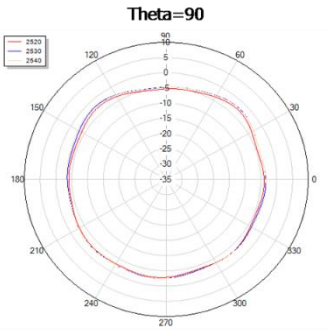
710 MHz		830 MHz	
			
			
900 MHz		**960 MHz**	
			
			

1710 MHz		1740 MHz	
			
			
1880 MHz		**1950 MHz**	
			
			

2140 MHz		2350 MHz	
			
			
2450 MHz		2600 MHz	
			
			

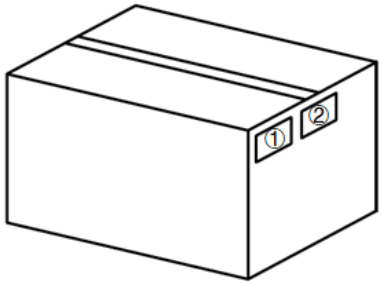
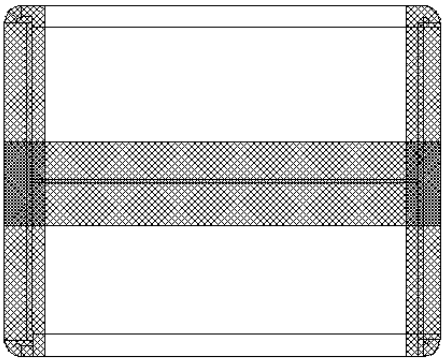
- **Max Peak Gain**

810 MHz		960 MHz	
			
			
2170 MHz		2320 MHz	
			
			

2400 MHz		2530 MHz	
			
			

4 Packaging

Step	Packaging Picture / 2D Picture	Description
1		10 products in a one-piece bag. (10 Antennas / PE Bag)
2		40 products in a PE bag. (40 Antennas / PE Bag)
3		(40 PE Bags / Carton Box) (1600 Antennas / Carton Box) Estimated quantity Products that cannot fill the entire carton box are packed in a suitable size carton box. Carton Size: L × W × H = 470 × 310 × 310 mm

4		**Position for Attaching Labels** ① Carton Label ② Quality Label
5		**Sealing Cartons** H-shaped sealing cartons
Note	The initial packaging method described above is for reference only, and the final actual packaging method shall be subject to the actual shipping packaging.	

Contact Us

At Quectel, our aim is to provide timely and comprehensive services to our customers. If you require any assistance, please contact our headquarters:

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Tel: +86 21 5108 6236
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Or our local offices. For more information, please visit:
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Revision History

Version	Date	Author	Note
-	2026-06-08	Ruean Liu/ Steven Mo/ Piers Ji/ Rainey Liao	Creation of the document
1.0	2026-06-08	Ruean Liu/ Steven Mo/ Piers Ji/ Rainey Liao	First official release



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